

Revision Notes for Class 8 Science Chapter 9: Solutes, Solvents and Solutions

When we mix substances together, they can form either a uniform mixture called a solution, or a non-uniform mixture. For example, when sugar and salt are dissolved in water, the resulting mixture is uniform—its taste remains the same throughout because the components are evenly distributed. In contrast, mixing chalk powder or sand with water does not result in a uniform mixture, as the particles remain unevenly distributed. Uniform mixtures, or solutions, are a key concept in this chapter.

Solute, Solvent, and Solution

A solution is formed when two or more substances mix to form a uniform mixture. In such a system, the solid component that dissolves is called the solute, and the liquid in which it dissolves is called the solvent. For example, in saltwater, salt is the solute, and water is the solvent. If two liquids are mixed and both dissolve, the one present in a smaller amount is called the solute and the other, the solvent. Interestingly, air also acts as a gaseous solution: nitrogen is considered the solvent as it is present in the largest amount, while oxygen, carbon dioxide, and other gases are the solutes.

Even when a solid is present in a larger quantity—as in the sugar syrup (chashni) of *Gulab jamun*—water remains the solvent, and sugar is still seen as the solute. This rule helps us clearly define what is dissolving and what is being dissolved in most real-life mixtures.

Saturated and Unsaturated Solutions

The amount of solute that dissolves in a fixed quantity of solvent varies depending on the substance and the temperature. When you add salt or sugar to water, at first it dissolves completely. If you keep adding more, eventually no more dissolves and some settles at the bottom. At this stage, the mixture is called a saturated solution. Before this point, when all added solute is dissolving, the solution is unsaturated. The amount of

solute that actually dissolves to form a saturated solution at a specific temperature is called the solubility of the solute.

The concentration of a solution indicates how much solute is present in a certain amount of solution or solvent. Solutions can be dilute (less solute) or concentrated (more solute), and these terms are relative—they compare the amount of solute, not absolute values.

Temperature and Solubility

Temperature strongly affects solubility. For most solids, increasing the temperature lets more solute dissolve in a solvent—so hot water can dissolve more sugar or salt than cold water. This property is especially relevant in kitchens and laboratories. For gases, however, solubility behaves in the opposite way: as water temperature rises, gases like oxygen dissolve less easily. That's why cold water in rivers and aquariums holds more oxygen for fish and aquatic plants.

- Solubility of solids generally increases with temperature.
- Solubility of gases generally decreases with temperature.

Mass, Volume, and Devices

Mass is the amount of matter in an object and is measured in grams (g) or kilograms (kg) using devices like a digital weighing balance or a two-pan balance. Volume is the amount of space an object occupies, measured in cubic metres (m^3), cubic centimetres (cm^3), litres (L), or millilitres (mL). For measuring the volume of liquids, a measuring cylinder is commonly used, which comes in different sizes and helps measure volumes accurately.

- $1 \text{ mL} = 1 \text{ cm}^3$ for water
- $1000 \text{ mL} = 1 \text{ L}$

- Mass and volume can be measured precisely using digital balances and graduated cylinders

To measure the mass of a solid, set the balance to zero (tare), place a weigh dish, tare again, and then place the object for the direct reading in grams. For measuring liquid mass, use a beaker instead of a dish. Remember, while mass refers to the amount of substance, weight is the force of gravity acting on that substance—yet most balances are marked in grams or kilograms.

Measuring Volume of Solids

For regular shapes like cubes or cuboids, volume is calculated by multiplying length, width, and height (Volume = $l \times w \times h$). For irregular solids like stones, volume is measured using the displacement method—immerse the object in a measuring cylinder containing water. The increase in water level equals the volume of the object. For example, if the water rises from 50 mL to 55 mL after lowering a stone, the stone's volume is 5 cm³.

Density: Key to Floating and Sinking

Density describes how much mass is packed into a particular volume, calculated as density = mass/volume. For example, the density of water at room temperature is about 1 g/cm³. If a substance's density is higher than the liquid it is placed in, it will sink; if lower, it will float. This explains why oil floats on water and why some objects, like wooden logs or bamboo, float while others sink.

- Density = Mass/Volume (SI unit: kg/m³ or g/cm³)
- Relative density compares a substance's density with that of water. It has no unit.
- If relative density > 1, the object sinks in water; if < 1, it floats.

To calculate relative density:

Relative density = (Density of substance) / (Density of water at the same temperature)

Effect of Temperature and Pressure on Density

Generally, heating a substance expands its volume (without changing its mass), so density decreases with temperature. That's why hot air rises: it is less dense than cool air, making hot air balloons work. For pressure, gases are much more affected than liquids or solids—compressing a gas increases its density, while liquids and solids are nearly incompressible.

An important exception is water. Its density is maximum at 4°C . As water freezes and turns to ice, it expands and its density decreases—so ice floats on liquid water. This property is crucial in nature, as it allows aquatic life to survive under the ice during cold weather.

Practice Problems and Applications

Classroom questions involve identifying true or false statements about solubility, solutions, and density, calculating densities from given mass and volume, and predicting if an object will float or sink. You may be asked to analyse why an unpeeled orange floats but a peeled one sinks, showing how air pockets and density affect floating. There are also practical topics, like why the Dead Sea supports almost no aquatic life—because its density is far higher due to the massive salt content.

Quick Recap

- Solutions are uniform mixtures of solutes (dissolved substances) and solvents (substances in which they dissolve).
- Saturated solutions can't dissolve more solute at a given temperature; unsaturated solutions can.
- Solubility and density are affected by temperature and, for gases mainly, by pressure.
- Density governs if things float or sink—and is easy to calculate if you measure both mass and volume accurately.

Traditional use of solvents can be seen in Indian medicine, food, and daily life—salt production, herbal extracts, and even examples in transport, like bamboo rafts and wooden boats. Learning about solutes, solvents, and solutions helps us understand many aspects of science and our world.

